

## Performance Testing Report

|                      |                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------|
| <b>Date</b>          | 15 March 2026                                                                                       |
| <b>Team ID</b>       | Regala Rakesh, Sri Dhruthi Mallela, P L P D Sravanthi Murukuri, Manikanta Cherukuri, Thokala Charan |
| <b>Project Name</b>  | EcoTrack - Personal Carbon AI Footprint Tracker                                                     |
| <b>Maximum Marks</b> | 5 Marks                                                                                             |

### Performance Testing Report:

Performance testing is critical to ensure your solution is fast, scalable, and stable under specific workloads. Document your testing strategy, the tools utilized, and the specific metrics measured against your target thresholds.

### Step 1: Test Scenarios & Tooling

| S.No | Test Type         | Scenario / API Endpoint Tested                                         | Tool(s) Used                          |
|------|-------------------|------------------------------------------------------------------------|---------------------------------------|
| 1    | Load Testing      | <i>[e.g., Simulating 500 concurrent users logging in]</i>              | <i>[e.g., Apache JMeter, Postman]</i> |
| 2    | Stress Testing    | <i>[e.g., Pushing the database with 10,000 queries/sec]</i>            |                                       |
| 3    | Spike Testing     | <i>[e.g., Sudden burst of traffic on the checkout page]</i>            |                                       |
| 4    | Endurance Testing | <i>[e.g., Running average load for 24 hours to check memory leaks]</i> |                                       |

### Step 2: Key Performance Indicators (KPIs) & Results

| S.No | Metric (KPI)              | Target Threshold                   | Actual Result         | Status      |
|------|---------------------------|------------------------------------|-----------------------|-------------|
| 1    | Average Response Time     | <i>[e.g., &lt; 200 ms]</i>         | <i>[e.g., 145 ms]</i> | <b>Pass</b> |
| 2    | Peak Response Time        | <i>[e.g., &lt; 500 ms]</i>         |                       |             |
| 3    | Throughput (Requests/sec) | <i>[e.g., &gt; 1000 RPS]</i>       |                       |             |
| 4    | Error Rate                | <i>[e.g., &lt; 1%]</i>             |                       |             |
| 5    | CPU/Memory Utilization    | <i>[e.g., &lt; 80% under load]</i> |                       |             |